Land 5: Osseous Peaks

Bone tissue and the skeleton · OpenStax Ch. 6 to 8

Key ideas

Functions and Bone Shapes

- Six functions: support, protection, movement (levers for muscles), mineral storage, fat storage, and hematopoiesis.
- Bone stores about 99% of the body's calcium and much of its phosphate.
- Long bones are longer than wide (femur, humerus, phalanges); short bones are cube shaped (carpals, tarsals).
- Flat bones are thin and often curved (sternum, ribs, cranial bones); irregular bones have complex shapes (vertebrae, facial bones).
- Sesamoid bones form within tendons; the patella is the main example.

Long Bone Anatomy and Marrow

- Diaphysis is the shaft around the medullary cavity; epiphyses are the wider ends filled with spongy bone.
- The metaphysis joins diaphysis and epiphysis and holds the epiphyseal plate, which becomes the epiphyseal line in adults.
- Periosteum covers the outer surface and is anchored by perforating (Sharpey) fibers; endosteum lines the medullary cavity.
- Hyaline articular cartilage covers joint surfaces of the epiphyses to reduce friction.
- Red marrow makes blood cells; yellow marrow stores fat. In adults red marrow is mainly in flat bones, vertebrae, hip bones, and proximal femur and humerus.

Bone Cells and Bone Tissue

- Osteogenic cells are stem cells that divide and become osteoblasts.
- Osteoblasts secrete osteoid and build matrix; osteocytes sit in lacunae and maintain matrix; osteoclasts are large multinucleated cells that resorb bone.
- Matrix is about one third collagen (flexibility) and two thirds calcium phosphate as hydroxyapatite (hardness).
- Compact bone is built of osteons: a central canal with vessels and nerves, rings of concentric lamellae, lacunae, and canaliculi.
- Perforating canals run at right angles and connect central canals to the periosteum and marrow.
- Spongy bone has trabeculae aligned along stress lines, is light, and holds red marrow; its osteocytes are fed from marrow vessels through canaliculi.

Ossification and Growth

- Intramembranous ossification turns mesenchyme directly into bone: flat skull bones, facial bones, mandible, and clavicles.
- Endochondral ossification replaces a hyaline cartilage model: primary ossification center in the diaphysis, secondary centers in the epiphyses.
- Epiphyseal plate zones from epiphysis toward diaphysis: reserve, proliferative, maturation and hypertrophy, calcified matrix.
- Growth in length happens at the epiphyseal plate; growth in width is appositional growth beneath the periosteum.
- Plates close in late adolescence to early adulthood, leaving an epiphyseal line; injuries through the plate can shorten a limb.

Remodeling, Nutrition, and Hormones

- Remodeling balances osteoclast resorption with osteoblast deposition throughout life; loading increases density (Wolff's law).
- Calcium and phosphorus build matrix; vitamin D enables calcium absorption; vitamin C is needed for collagen; vitamin K supports bone proteins.
- Growth hormone drives growth at the plates; thyroxine supports it; estrogen and testosterone cause the pubertal spurt and then close the plates.
- Excess growth hormone before plate closure causes gigantism; after closure it causes acromegaly with thickened bones.

Calcium Homeostasis

- Low blood calcium triggers PTH from the parathyroid glands.
- PTH stimulates osteoclasts, increases kidney calcium reabsorption, and stimulates calcitriol production by the kidneys.
- Calcitriol, the active form of vitamin D, increases calcium absorption in the small intestine.
- High blood calcium triggers calcitonin from thyroid parafollicular (C) cells, which inhibits osteoclasts.
- Low calcium raises nerve and muscle excitability, causing spasms and tetany; high calcium depresses excitability.

Fractures and Disorders

- Open fractures pierce skin; closed fractures do not. Transverse, oblique, spiral, comminuted, impacted, greenstick, and compression are common types.
- Repair stages: fracture hematoma, fibrocartilaginous callus, bony callus of spongy bone, then remodeling.
- Osteoporosis: resorption exceeds deposition, making bones porous; common after menopause as estrogen falls; causes hip, wrist, and vertebral compression fractures.
- Rickets in children and osteomalacia in adults result from too little vitamin D or calcium, leaving soft bones.
- Greenstick fractures are incomplete breaks typical of flexible bones in children.

Axial Skeleton

- Axial skeleton has 80 bones: skull, hyoid, auditory ossicles, vertebral column, and thoracic cage; the appendicular skeleton has 126.
- Skull: 8 cranial and 14 facial bones joined by sutures (coronal, sagittal, lambdoid, squamous); occipital bone holds the foramen magnum.
- Fontanelles are soft spots in the infant skull; the anterior fontanelle lies between frontal and parietal bones.
- Paranasal sinuses are in the frontal, ethmoid, sphenoid, and maxillary bones; the hyoid articulates with no other bone.
- Vertebrae: 7 cervical, 12 thoracic, 5 lumbar, plus sacrum and coccyx. Thoracic and sacrococcygeal curves are primary; cervical and lumbar are secondary.
- Atlas (C1) has no body; axis (C2) has the dens. Ribs 1 to 7 are true, 8 to 12 false, 11 and 12 floating. Sternum: manubrium, body, xiphoid process.

Appendicular Skeleton

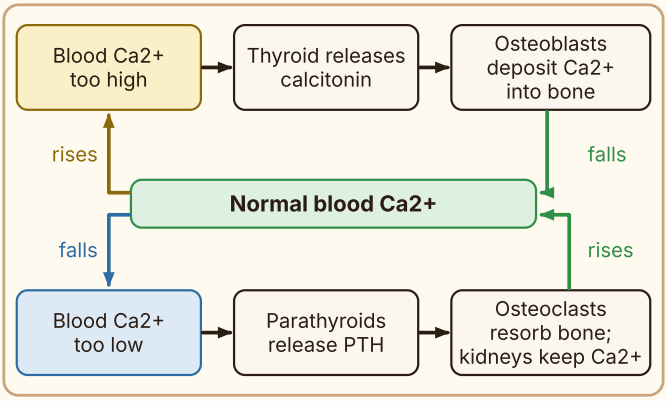
- Pectoral girdle: clavicle and scapula. Humerus has a surgical neck below the tubercles, a common fracture site.
- Radius is lateral (thumb side); ulna is medial and its trochlear notch grips the humerus. There are 8 carpals; the scaphoid is fractured most often.
- Hip bone forms from fused ilium, ischium, and pubis meeting at the acetabulum; the female pelvis has a wider, rounder inlet and a wider subpubic angle.
- Femur is the longest, strongest bone; the patella is a sesamoid bone; the tibia bears weight and forms the medial malleolus; the fibula forms the lateral malleolus.
- There are 7 tarsals; the talus forms the ankle joint and the calcaneus forms the heel. Longitudinal (medial and lateral) and transverse arches distribute weight.

Villager wisdom

- Osteoblasts build, osteoclasts carve, and osteocytes keep watch from their lacunae. Remember B for build and C for carve.
- When blood calcium drops, the parathyroid glands send PTH: osteoclasts free calcium from bone and the kidneys hold on to it. Calcitonin does the reverse.
- Bones are more than scaffolding. They protect organs, anchor muscles for movement, store calcium and fat, and their red marrow makes blood cells.
- A broken bone heals in four stages: hematoma, fibrocartilaginous callus, bony callus, and finally remodeling back toward its old shape.
- The axial skeleton is the body's central pillar of skull, spine, and rib cage. The limbs and the girdles that hold them make up the appendicular skeleton.

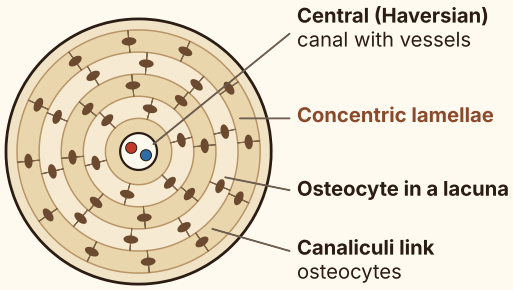
Diagrams

Calcium homeostasis



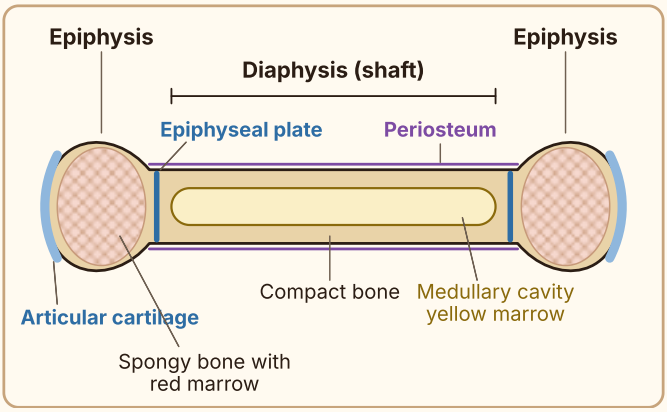
PTH raises blood calcium (osteoclasts resorb bone, kidneys keep calcium and make calcitriol); calcitonin lowers it as osteoblasts deposit calcium in bone.

The osteon



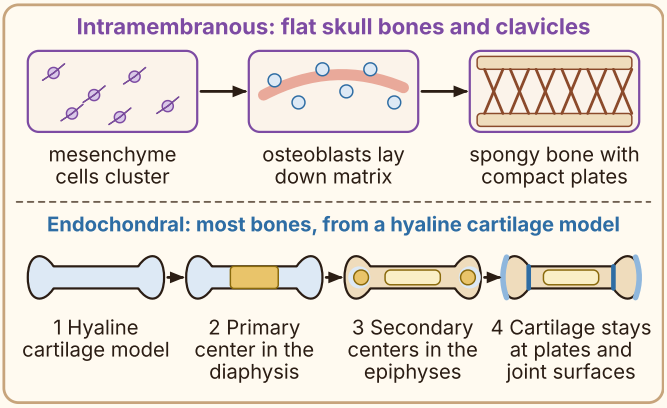
Compact bone is built of osteons: rings of matrix (lamellae) around a central canal, with osteocytes in lacunae linked by tiny canaliculi.

Anatomy of a long bone



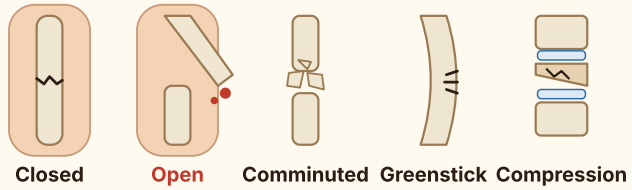
A long bone grows in length at the epiphyseal plates; its hollow diaphysis of compact bone surrounds the medullary cavity, while spongy bone fills the epiphyses.

Intramembranous and endochondral ossification



Flat skull bones and the clavicles form directly from mesenchyme (intramembranous); most other bones replace a hyaline cartilage model (endochondral), starting at a primary center in the diaphysis.

Fracture types and repair

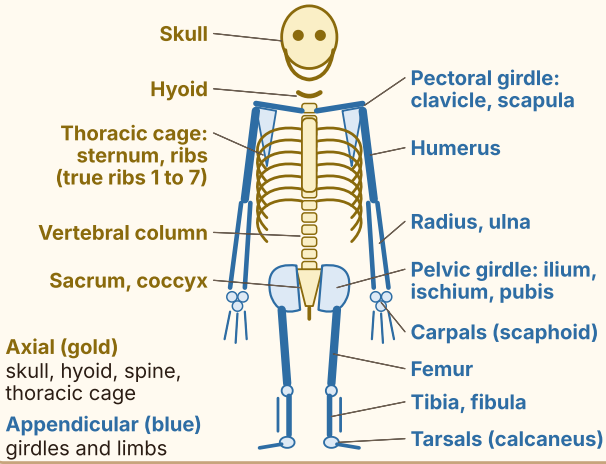


Fracture repair

1 Hematoma: clot fills the break → 2 Soft callus: fibrocartilage bridges it → 3 Bony callus: spongy bone replaces it → 4 Remodeling: extra bone removed

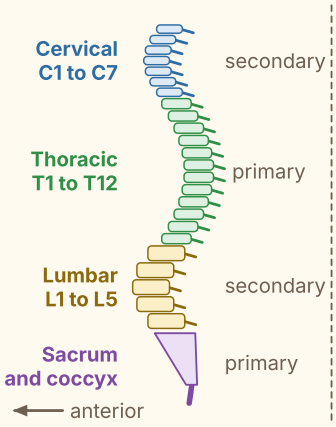
Fractures are named by how the bone breaks; repair runs from hematoma to a fibrocartilaginous callus, then a bony callus that remodeling slowly smooths away.

Axial and appendicular skeleton

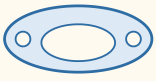


The axial skeleton is the skull, hyoid, vertebral column and thoracic cage; the appendicular skeleton is the pectoral and pelvic girdles plus the limbs attached to them.

The vertebral column



Atlas (C1), from above



no body and no spinous process

Cervical vertebra

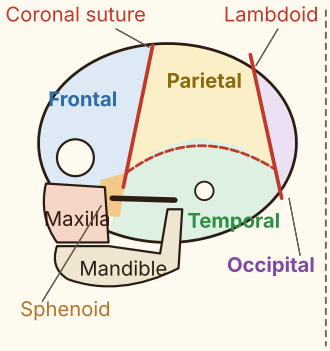


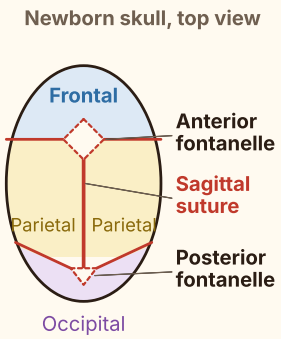
transverse foramina carry the vertebral arteries

Lordosis: exaggerated lumbar curve (late pregnancy)
Kyphosis: thoracic hump

There are 7 cervical, 12 thoracic and 5 lumbar vertebrae; the thoracic and sacral curves are primary (present at birth), while the cervical and lumbar curves are secondary.

Bones of the skull

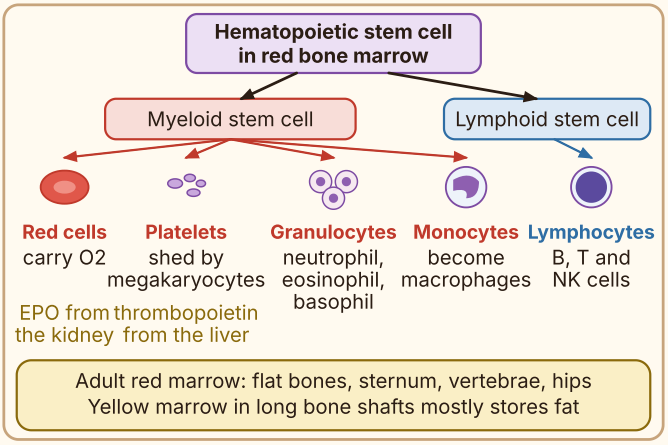




Paranasal sinuses: frontal, ethmoid, sphenoid and maxilla
Foramen magnum (occipital bone): spinal cord meets brainstem

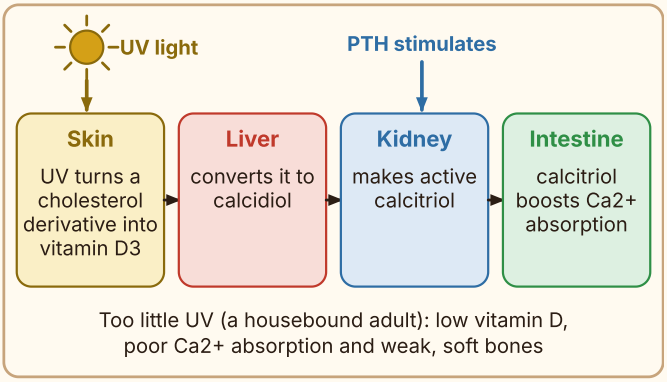
The frontal, parietal, temporal and occipital bones meet at sutures; in a newborn the anterior fontanelle lies where the frontal and parietal bones meet, and the sagittal suture joins the two parietals.

Hematopoiesis



All formed elements come from hematopoietic stem cells in red bone marrow; erythropoietin from the kidney drives red cell production and thrombopoietin from the liver drives platelet production.

Vitamin D synthesis



UV light starts vitamin D synthesis in the skin, the liver modifies it, and the kidney makes active calcitriol, which lets the intestine absorb calcium for strong bones.

Self check

1. **Bone marrow samples from adults are usually taken from the iliac crest rather than the femoral shaft. Why?**
    A. The femoral shaft has no medullary cavity
    B. The iliac crest contains only yellow marrow
    C. The femur lacks blood vessels in adults
    D. Adult red marrow is mainly in flat bones and hip bones
2. **Weeks after a tibia fracture, a firm bulge can be felt at the break. Months later the bulge is nearly gone. What explains the change?**
    A. Remodeling of the bony callus
    B. Dissolving of the fracture hematoma
    C. Formation of a fibrocartilaginous callus
    D. Growth at the epiphyseal plate
3. **Why is spongy bone well suited to the ends of long bones?**
    A. It has the most densely packed osteons
    B. It resists stress from many directions yet is light
    C. It contains no red marrow
    D. It lacks osteocytes
4. **Which feature makes the atlas (C1) different from all other vertebrae?**
    A. It has a dens projecting upward
    B. It has no transverse foramina
    C. It has no body and no spinous process
    D. It has facets for rib articulation
5. **What is a fracture called when the broken bone pierces through the skin?**
    A. Open (compound) fracture
    B. Closed (simple) fracture
    C. Greenstick fracture
    D. Transverse fracture
6. **Which bone of the axial skeleton does not articulate with any other bone?**
    A. Vomer
    B. Mandible
    C. Coccyx
    D. Hyoid bone
7. **Which suture joins the two parietal bones along the midline of the skull?**
    A. Coronal suture
    B. Sagittal suture
    C. Lambdoid suture
    D. Squamous suture

8. **During thyroid surgery, the parathyroid glands are removed by accident. What is the likely early problem?**
    A. High blood calcium
    B. Low blood calcium with muscle spasms
    C. Rapid growth in bone length
    D. Bone loss from excess calcitonin
9. **What happens at the epiphyseal plate that allows a long bone to grow in length?**
    A. Osteoclasts add bone under the periosteum
    B. Compact bone converts to spongy bone
    C. Marrow converts into cartilage
    D. Cartilage grows and is replaced by bone
10. **In compact bone, what are the tiny channels that connect lacunae to one another and to the central canal?**
    A. Perforating canals
    B. Central canals
    C. Trabeculae
    D. Canaliculi
11. **Which statement about bone remodeling is correct?**
    A. It continues throughout adult life
    B. It stops when the growth plates close
    C. It is carried out only by osteoblasts
    D. It occurs only after a fracture
12. **What is the basic structural unit of compact bone?**
    A. Trabecula
    B. Epiphyseal plate
    C. Periosteum
    D. Osteon (Haversian system)

Answer key

1. **D.** Adult red marrow is mainly in flat bones and hip bones. In adults, red marrow remains mainly in flat bones, vertebrae, the hip bones, and the proximal ends of the femur and humerus, while long bone shafts fill with yellow marrow. The femoral shaft does have a medullary cavity, but it holds mostly fat.
2. **A.** Remodeling of the bony callus. In the final stage of repair, osteoclasts and osteoblasts remodel the bony callus, removing excess bone and restoring the original shape. The hematoma is cleared in the first days, long before a firm bony bulge exists.
3. **B.** It resists stress from many directions yet is light. Trabeculae align along lines of stress and leave spaces, making bone lighter while still strong. Spongy bone actually houses red marrow, so the option saying it has none is wrong.
4. **C.** It has no body and no spinous process. The atlas is a ring of bone with anterior and posterior arches but no body or spinous process. The dens belongs to the axis (C2), and transverse foramina are present in all cervical vertebrae, including C1.
5. **A.** Open (compound) fracture. In an open fracture the bone breaks through the skin, which raises the risk of infection. In a closed fracture the skin remains intact.

6. **D.** Hyoid bone. The hyoid is suspended in the neck by muscles and ligaments and does not form a joint with another bone. The mandible articulates with the temporal bones at the temporomandibular joints.
7. **B.** Sagittal suture. The sagittal suture runs front to back between the right and left parietal bones. The coronal suture instead joins the frontal bone to the parietal bones.
8. **B.** Low blood calcium with muscle spasms. Without PTH, blood calcium falls, and low calcium makes nerves and muscles overly excitable, producing spasms called tetany. High blood calcium is what happens with too much PTH, not too little.
9. **D.** Cartilage grows and is replaced by bone. Chondrocytes multiply on the epiphyseal side while cartilage on the diaphyseal side calcifies and is replaced by bone. Growth in width, not length, comes from appositional growth beneath the periosteum.
10. **D.** Canaliculi. Canaliculi are microscopic channels that let osteocytes exchange nutrients and wastes and communicate through gap junctions. Perforating canals are much larger and carry blood vessels across the bone.
11. **A.** It continues throughout adult life. Osteoclasts and osteoblasts keep resorbing and depositing bone throughout life, replacing part of the skeleton every year. Closure of the growth plates ends lengthwise growth, not remodeling.
12. **D.** Osteon (Haversian system). An osteon is a set of concentric lamellae around a central canal that carries blood vessels and nerves. Trabeculae are the lattice units of spongy bone, not compact bone.

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